

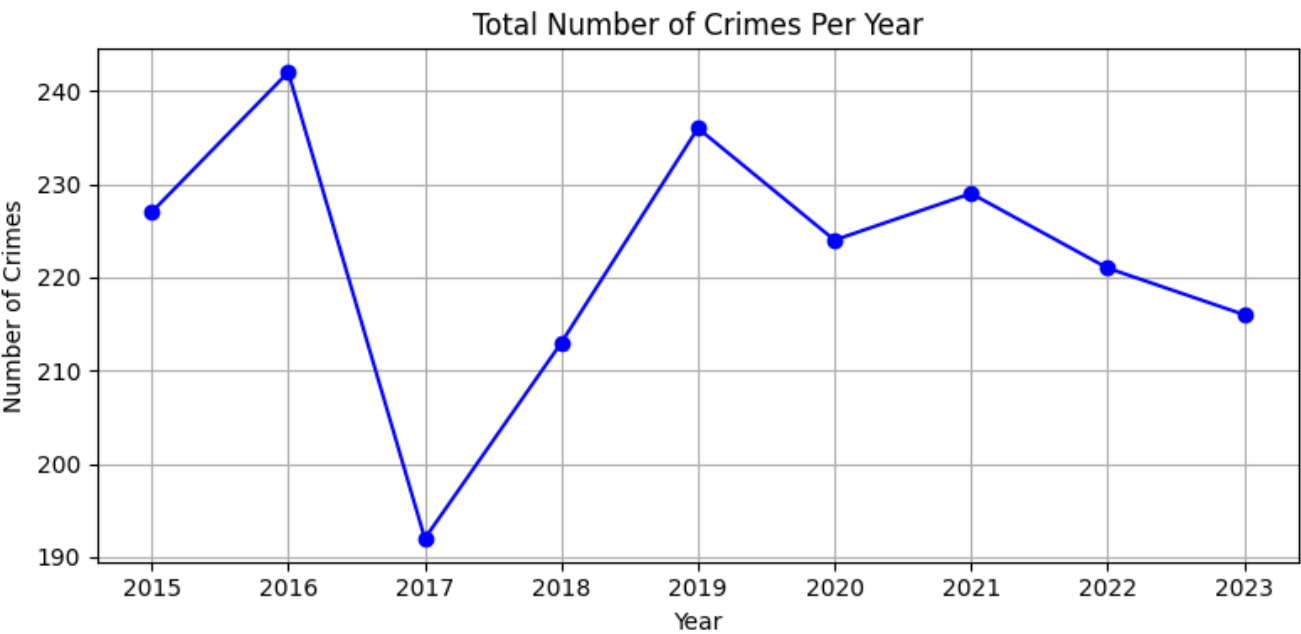
Use Case 2 — Exploratory Analysis & Visualization of Crime

Objective: Analyze trends and patterns using Pandas, NumPy, Matplotlib, and Seaborn.

Tasks

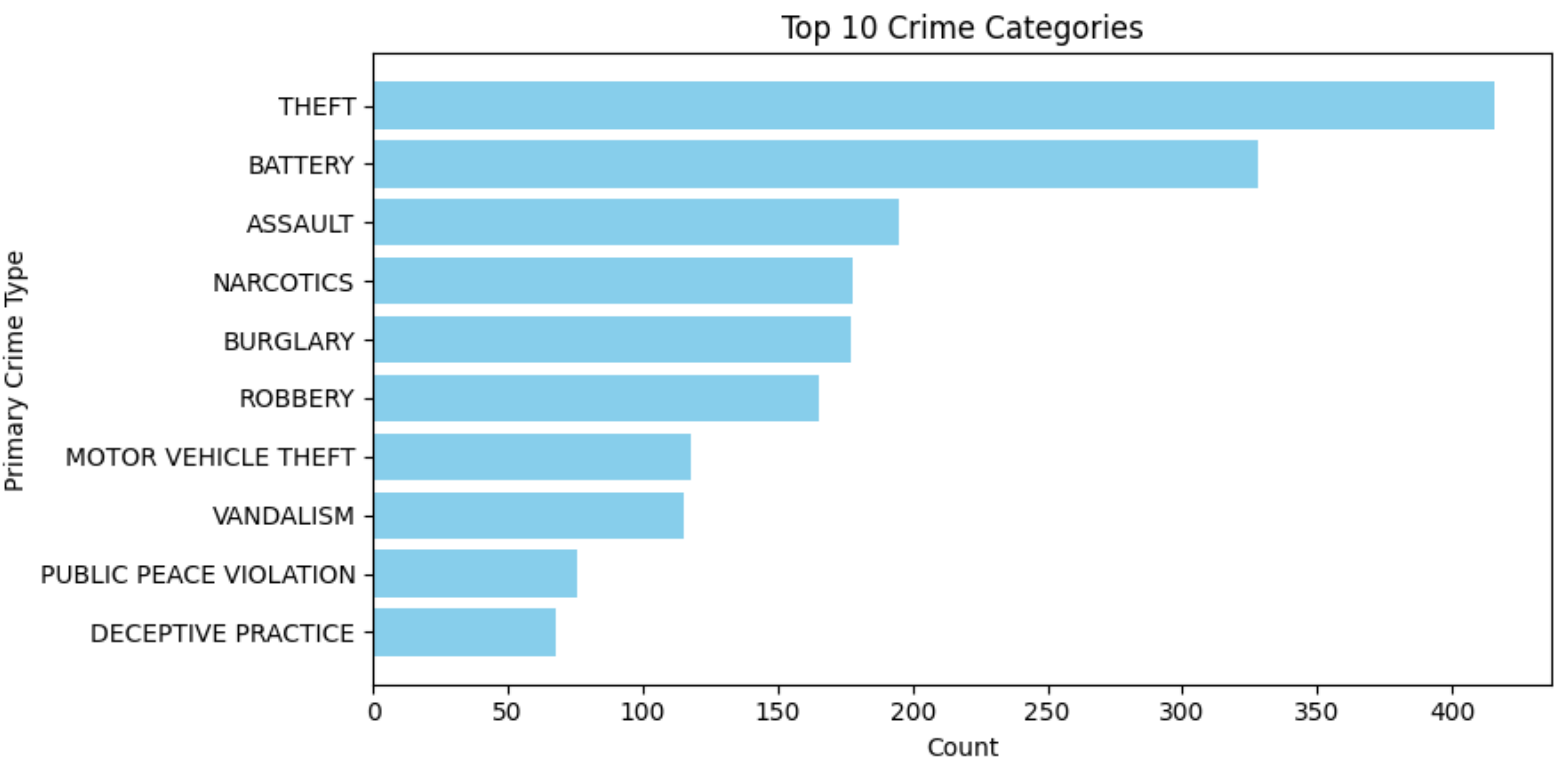
1. Crime Trend Over Years

- Plot total number of crimes per year.
- Visual interpretation: Is the crime rate rising or decreasing?



2. Crime Distribution by Category

- Bar chart of top 10 crime categories (Primary Type).
- Calculate counts and percentage for each.



3. Arrests and Crime Outcomes

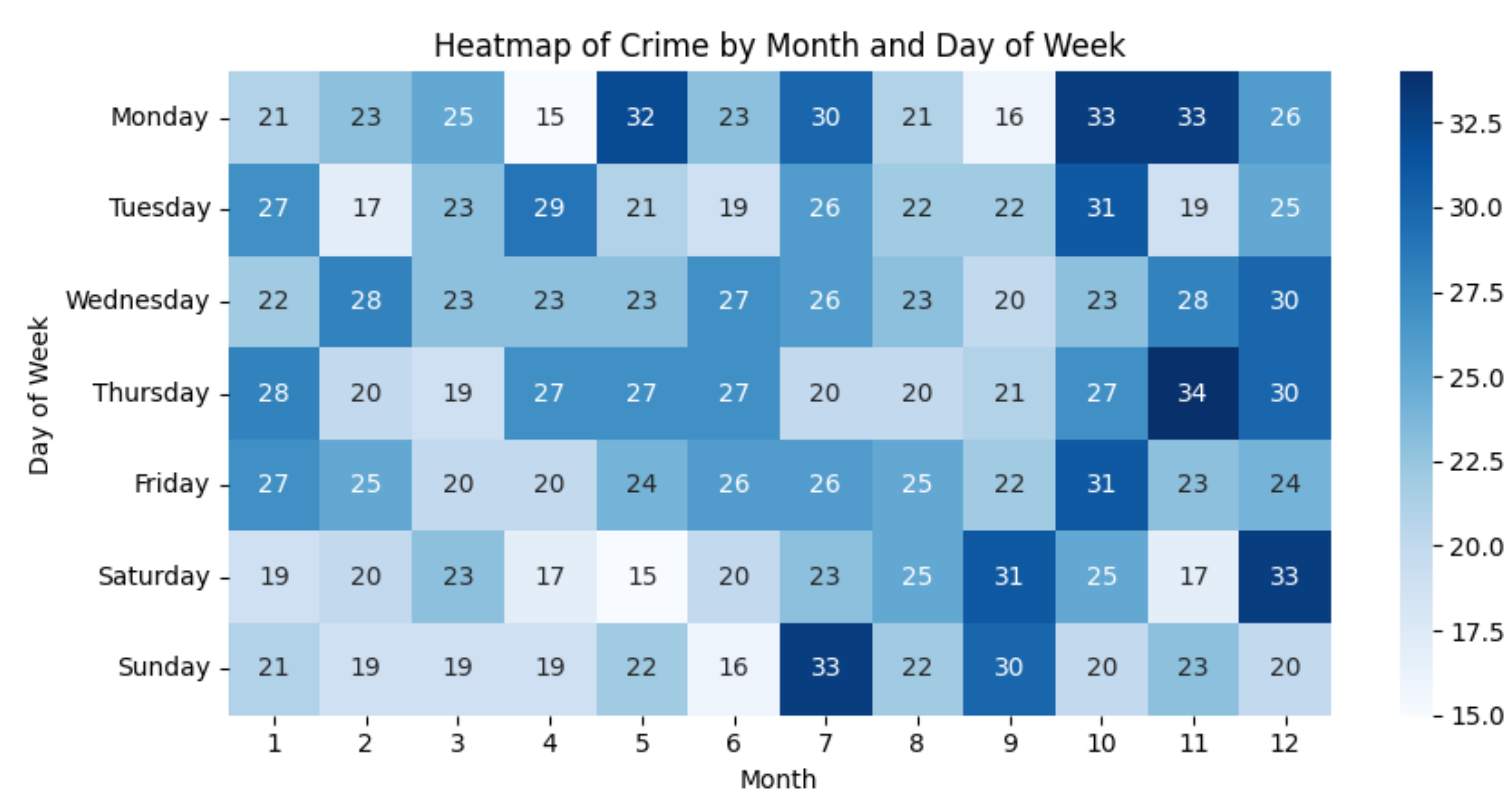
- What percentage of crimes result in arrest?
- `arrest_rate = df['Arrest'].mean() * 100`

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Top 10 Crime Categories:
      Primary_Type  Count  Percentage
0          THEFT    416     20.80
1        BATTERY    328     16.40
2        ASSAULT    195      9.75
3      NARCOTICS    178      8.90
4      BURGLARY    177      8.85
5        ROBBERY    165      8.25
6  MOTOR VEHICLE THEFT  118      5.90
7      VANDALISM    115      5.75
8  PUBLIC PEACE VIOLATION   76      3.80
9  DECEPTIVE PRACTICE    68      3.40

Overall Arrest Rate: 29.80%
Yearly Arrest Rates:
Year
2015    27.753304
2016    25.619835
2017    29.166667
2018    33.802817
2019    31.779661
2020    30.357143
2021    29.694323
2022    32.579186
2023    27.777778
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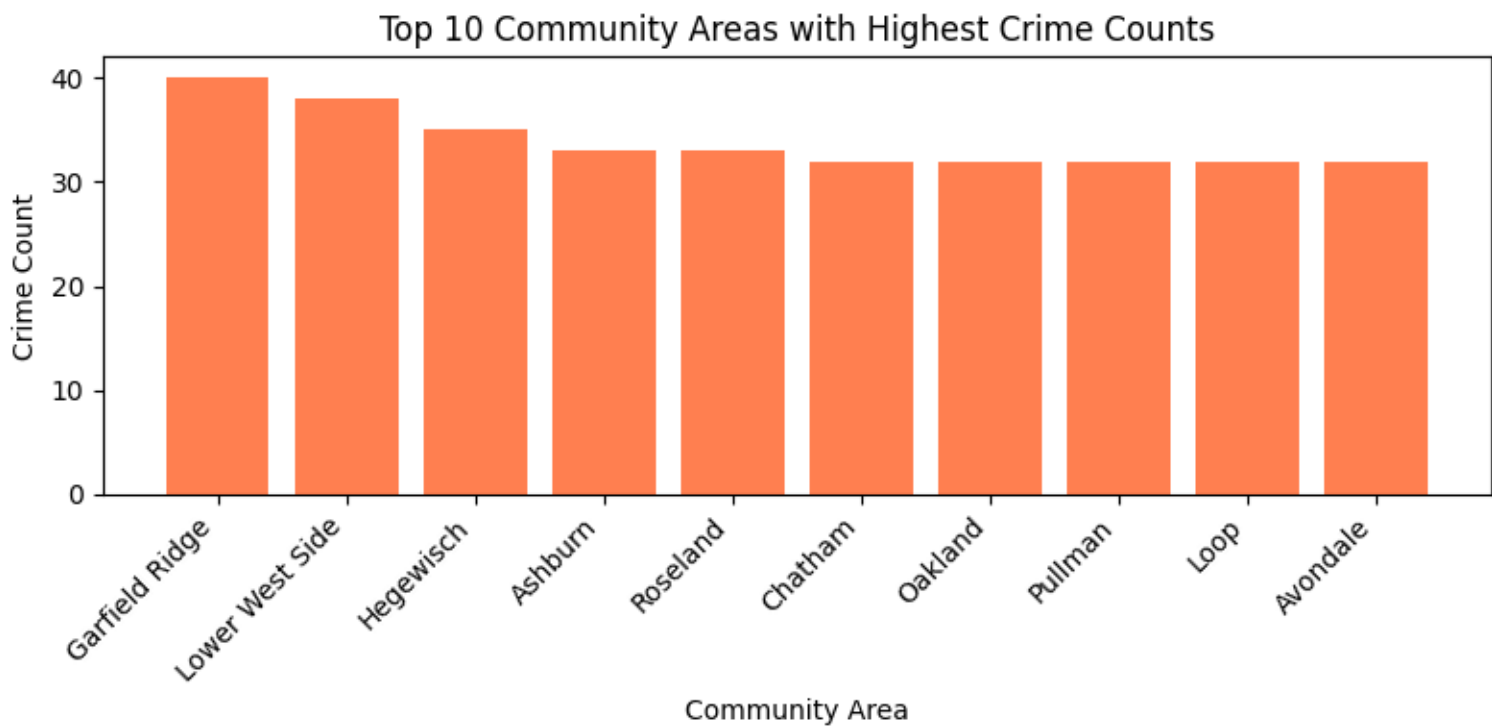
4. Heatmap of Crime by Month and Day of Week

- Use Seaborn heatmap of crime frequency pivoted by Month vs DayOfWeek.



5. Top Community Areas

- List top 10 community areas with highest crime counts.
- Plot on a bar chart.



Questions

- Which crime category is most frequent?
- Is the arrest rate consistent across different years?
- Which month has the highest crime frequency?

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--- Questions & Answers ---
1. Which crime category is most frequent? THEFT
2. Is the arrest rate consistent across different years? Yearly rates range from 25.62% to 33.80%.
3. Which month has the highest crime frequency? Month 10
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